

The background of the slide is a photograph of a field of tall, thin grasses and small yellow flowers. The grasses are in the foreground, some in sharp focus and others blurred. The flowers are scattered throughout the field, with a few in the upper right corner. The sky is a clear, pale blue. The overall lighting is soft and natural, suggesting a sunny day.

Prompt Engineering

Dr. Kornelia Papp

Agenda

01 Session goals

02 Prompt engineering

03 Exercise

04 Prompt techniques

05 Quiz

Session Goals

In this session, we learn how to guide LLMs more reliably by improving the task definition, context, and output contract.

What is prompt engineering?

Understand prompting as the control surface for generative AI: not just asking questions, but specifying task, context, evidence, and expected output. Explore how temperature, Top P, length, stop sequences, and penalties affect creativity, consistency, verbosity, and control.

Why does prompt quality matter?

See how better prompts improve relevance, structure, safety, cost, latency, and reliability before changing the model or architecture.

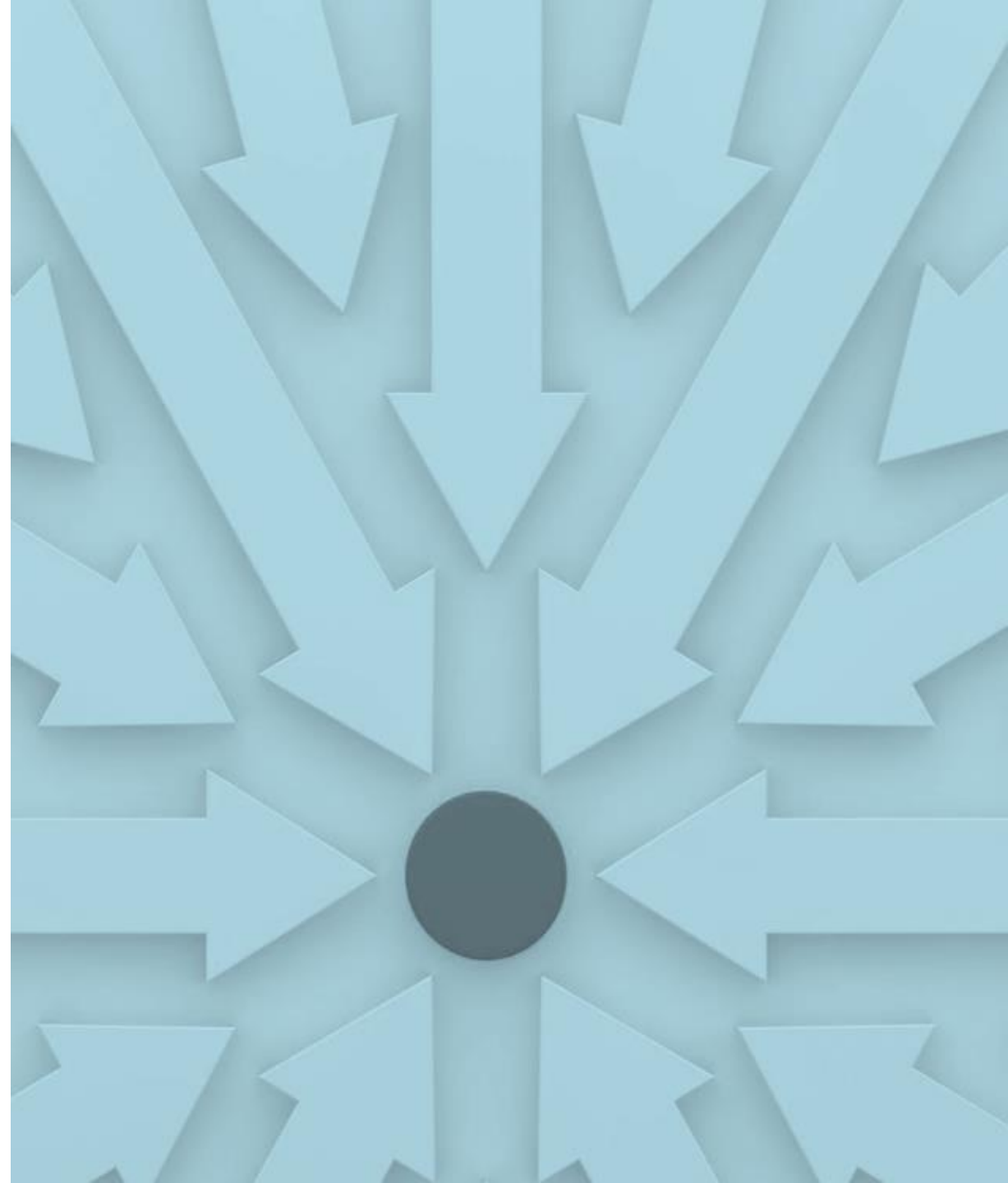
What makes a prompt effective?

Learn the core components of good prompts: instruction, context, input data, and output format.

Which prompting techniques should we use when?

Compare zero-shot, few-shot, meta-prompting, prompt chaining, Reflexion, least-to-most prompting, and structured output contracts.

Prompt
engineering turns
vague intent into
reliable AI
behaviour.



Agenda

01

Session goals

02

Prompt engineering

03

Exercise

04

Prompt techniques

05

Quiz

What is Prompt Engineering

Controlling generative AI systems with better specifications, evidence, and output contracts.

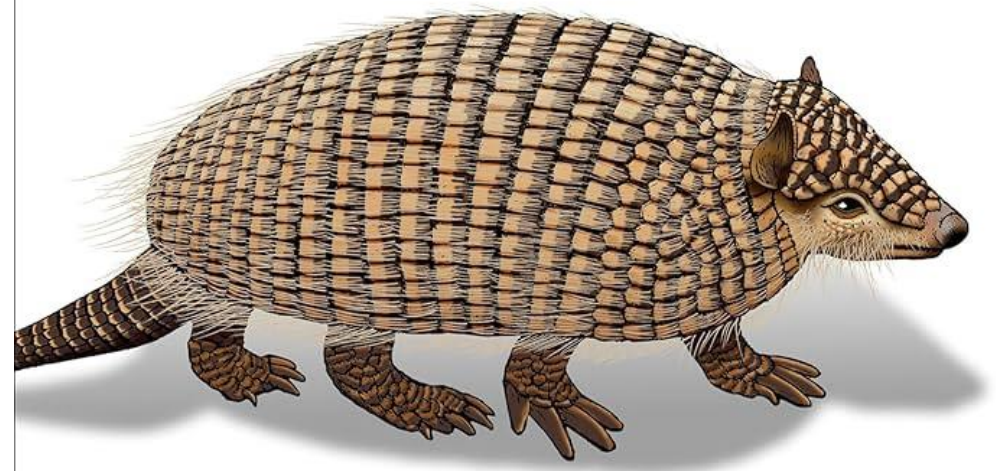
Prompting is also an **experimental** design practice: prompts are hypotheses that should be tested, compared, and iterated.

Suggested reading: <https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/>

O'REILLY®

Prompt Engineering for Generative AI

Future-Proof Inputs for Reliable AI Outputs



James Phoenix
& Mike Taylor

Why Prompt Engineering Matters

Even with stronger models, the quality of the specification still changes the answer you get.

Quality

better relevance, depth, and structure

Safety

fewer unsupported or risky completions

Economics

less iteration, lower latency, lower cost

Before you change the model or architecture, improve the task definition, the context, and the output contract.

A prompt is not just a question. It is the control surface for a stochastic system.

Why Prompt Engineering Matters Across Roles

Persona

Role of prompt engineering

Developer

Supports the design of robust interfaces between users, LLMs, tools, APIs, and workflows.

Product manager

Turns vague AI potential into usable product behaviour that can be tested, improved, and scaled.

Business leader

Builds a realistic understanding of what LLMs can do, where human oversight is needed, and how to create value safely.

Knowledge worker

Improves everyday tasks such as summarizing, drafting, comparing, analysing, and planning.

Risk or compliance expert

Helps expose limitations, reduce unsafe outputs, and design more controlled AI interactions.

Data scientist

Reveals how LLMs respond to different instructions, formats, examples, and constraints.

LLM Behaviour

When testing prompts via an API, model behaviour can be tuned with a few parameters. These settings help balance **accuracy, creativity, structure, cost, and reliability**.

Setting	What it controls	Typical use
Temperature	Randomness of the output	Low for factual QA; higher for creative writing
Top-p	Range of token choices considered	Low for focused answers; higher for diversity
Max Length	Maximum output token	Controls verbosity and cost
Stop Sequences	Where generation should stop	Enforces structure, lists, formats, limits
Frequency Penalty	Reduces repeated words	Useful when outputs become repetitive
Presence Penalty	Encourages new topics/phrases	Useful for more varied or creative text

Practical guidance:

- Adjust temperature or top-p, not both.
- Adjust frequency or presence penalty, not both.
- Test settings experimentally, because results vary by model and version.

LLM Behaviour

Try: same model, different settings

Fire and Ice

Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.

BY ROBERT FROST

Explain this poem:
Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.

Model: GPT-4.1 nano
Temperature: 0

 GPT-4.1 nano Hobby

Synced    + ...

 Explain this poem: Some say the world will end in fire, Some say in ice. From what I've tasted of desire I hold with those who favor fire. But if it had to perish twice, I think I know enough of hate To say that for destruction ice Is also great And would suffice.

Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.



 This poem is "Fire and Ice" by Robert Frost. It explores the themes of destruction and the possible ways the world might end. The speaker reflects on two powerful human emotions—desire (fire) and hate (ice)—and uses them as metaphors for destructive forces.

- The lines "Some say the world will end in fire, / Some say in ice" suggest different visions of destruction: fiery passion or cold indifference.
- The speaker personally aligns with fire, associating it with desire, which can be passionate and consuming.
- However, the poem also considers that if the world were to end a second time, ice—symbolizing hate, coldness, and emotional detachment—would be equally capable of causing destruction.
- The concluding lines imply that both intense desire and intense hate are destructive forces, capable of bringing about the end of the world.

Overall, the poem reflects on the destructive potential of human emotions and the idea that both passionate love and cold hatred can lead to ruin.



Model: GPT-4.1 nano
Temperature: 2

 GPT-4.1 nano Hobby

Synced    + ...

 Explain this poem: Some say the world will end in fire, Some say in ice. From what I've tasted of desire I hold with those who favor fire. But if it had to perish twice, I think I know enough of hate To say that for destruction ice Is also great And would suffice.

Some say the world will end in fire,
Some say in ice.
From what I've tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.



 This poem is "Fire and Ice" by Robert Frost. It explores the themes of destruction and the possible ways the world might end—either through fire or ice.

The "fire" symbolizes desire, passion, or intense emotion, which can lead to destruction through greed, anger, or obsession. The speaker admits to feeling this desire ("from what I've tasted of desire") and aligns with the idea that fire could bring about the world's end.

The "ice" represents hate, coldness, or indifference—emotions that can also cause destruction, but through a different means—coldness, apathy, or cruelty. The speaker suggests that if the world were to perish a second time, hate and coldness could be just as effective in ending it.

Overall, the poem uses metaphorical imagery to reflect on human emotions—desire and hate—and their destructive potential. It suggests that both passionate and cold, indifferent feelings have the power to bring about catastrophe, emphasizing the destructive capacity inherent in human nature.



Prompt Design: Components

A good prompt often combines four building blocks:

Component

Purpose

Instruction

What the model should do

Context

Background information that guides the answer

Input data

The text, question, or data to process

Output contract

The expected format or answer type

Prompt Design

You are a literature teacher explaining poetry to students who are new to literary analysis.

Task:

Explain the poem below in a clear and accessible way.

Poem:

“Some say the world will end in fire,
Some say in ice.
From what I’ve tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.”

Please structure your answer as follows:

1. Literal meaning

Explain what the poem says on the surface.

2. Symbolic meaning

Explain what “fire” and “ice” may represent.

3. Main idea

What is the poem really saying about human emotions?

4. Tone and mood

Describe the feeling of the poem in simple words.

5. Short takeaway

End with one sentence that captures the poem’s message.

Keep the explanation short, clear, and suitable for a classroom discussion.

Model: GPT-4.1 nano
Temperature: 0.70
Top P: 1

▲ Sure! Here's a simple and clear explanation of the poem:

1.

Literal meaning: The poem talks about two ideas for how the world might end — in fire or in ice. The speaker says they believe fire, which symbolizes desire, could cause destruction. But they also think ice, which symbolizes hate, could also lead to destruction if the world were to end twice.

2.

Symbolic meaning: "Fire" represents strong, passionate feelings like desire or greed. "Ice" stands for coldness, indifference, or hatred. Both are powerful emotions that can hurt or destroy.

3.

Main idea: The poem suggests that intense human emotions, like desire and hate, can be dangerous and destructive. Both passion and coldness can threaten the world.

4.

Tone and mood: The tone is thoughtful and slightly serious. The mood feels a bit tense, as if warning us about the power of strong feelings.

5.

Short takeaway: The poem tells us that both passionate love and intense hate have the power to destroy everything.

This helps us see how emotions can be so strong that they might cause big problems.

Model: Gemini 2.0 Flash
Temperature: 0.70
Top P: 1

🌈 Alright class, let's take a look at this poem by Robert Frost. It's called "Fire and Ice."

1. Literal meaning:

On the surface, the poem is about how the world might end. Some people believe it will be destroyed by fire, and others think it will be destroyed by ice. The speaker says that based on their experience with desire, they agree with the people who think fire will end the world. However, if the world had to end a second time, the speaker knows enough about hate to say that ice would be just as effective at destroying it.

2. Symbolic meaning:

Here, "fire" and "ice" aren't just about literal fire and ice. They represent powerful human emotions. "Fire" symbolizes passion, desire, and intense emotions that can consume us. "Ice," on the other hand, represents hatred, indifference, and coldness. Think of it like, fire is a quick burning rage, and ice is a slow, creeping resentment.

3. Main idea:

The poem is really about the destructive power of human emotions. It suggests that both intense passions like desire and colder, more calculating emotions like hate are capable of causing immense destruction. The poem argues that hate can be just as destructive as desire, even if it works in a different, slower way.

4. Tone and mood:

The poem has a thoughtful and somewhat cynical tone. There's a sense of world-weariness, as if the speaker has seen enough of both desire and hate to understand their potential for destruction. The mood is a bit somber, but also reflective.

5. Short takeaway:

Our strongest emotions, both fiery passions and icy hatred, have the potential to destroy us and the world around us.

Agenda

01

Session goals

02

Prompt engineering

03

Exercise

04

Prompt techniques

05

Quiz



Explore & Discuss

- Go to Vercel Playground: <https://ai-sdk.dev/playground>

Try:

- Same model, simple prompt
- Same prompt, different models
- Structured prompt with components, different models

Exchange observations and lessons learned with your classmates. What changed, what improved, and what still failed?

Prepare one observation about model differences and one observation about prompt structure.

Inspiration

Experiment

Same model + vague prompt

Same model + structured prompt

Same prompt + different models

Same model + different roles

Same task + different audiences

Same task + different output formats

Same prompt + constraints

What to compare

What assumptions does the model make?

Does structure improve clarity and usefulness?

Which model follows instructions best?

Does “act as a teacher / lawyer / product manager” change the answer?

How does the answer change for a child, expert, manager, or customer?

Bullet points vs table vs paragraph vs checklist

What happens when you add word limit, tone, structure, or “do not invent facts”?

Agenda

01

Session goals

02

Prompt engineering

03

Exercise

04

Prompt techniques

05

Quiz

Prompting Techniques

Zero-Shot Prompting: Giving a direct instruction with no examples. It relies entirely on the model's pre-trained knowledge.

Best for simple tasks like "Summarize this text" or "Translate this to French."

Few-Shot Prompting: Providing 2-5 examples of the input and desired output within the prompt. This "primes" the model to follow a specific pattern or tone.

Best for complex formatting (like JSON), mimicking a specific writing style, or classification tasks.

Prompt chaining: Decompose a complex task into smaller connected prompts. Each step produces an intermediate result that becomes the input for the next step, making the workflow more reliable, transparent, and easier to control.

Zero-shot Prompting

Zero-shot prompting refers to the use of a large language model to perform a task without providing task-specific examples in the prompt. Instead, the user specifies the desired task through instructions, relying on the model's pretraining and instruction-tuning to generate an appropriate response.

Classify the sentiment of the sentence below.

Sentence: "The dish was interesting."

Choose exactly one label: Positive, Neutral, or Negative.

Zero-shot Prompting



Classify the sentiment of the sentence below.

Sentence: "The dish was interesting."

Choose exactly one label: Positive, Neutral, or Negative.



Neutral

Zero-shot = instruction only, no examples.

It works well when the task is familiar to the model, such as summarizing, classifying, translating, rewriting, or explaining.

Few-shot Prompting

Few-shot prompting refers to a prompting strategy in which a small set of task-specific examples is included in the prompt before presenting a new instance to be solved. These examples serve as demonstrations that guide the model in inferring the intended task, expected output format, permissible labels or response categories, and the desired style of response.

Classify the sentiment as Positive, Neutral, or Negative.

Example 1: Sentence: "The food was excellent." Sentiment: Positive

Example 2: Sentence: "The food was terrible." Sentiment: Negative

Example 3: Sentence: "The food was okay." Sentiment: Neutral

Now classify: Sentence: "The dish was interesting." Sentiment:

Few-shot Prompting



Classify the sentiment as Positive, Neutral, or Negative.

Example 1: Sentence: "The food was excellent." Sentiment: Positive

Example 2: Sentence: "The food was terrible." Sentiment: Negative

Example 3: Sentence: "The food was okay." Sentiment: Neutral

Now classify: Sentence: "The dish was interesting." Sentiment:



Neutral

Few-shot prompting is useful when a zero-shot instruction is not enough. The examples show the model **what kind of answer is expected, which categories are allowed, and how the output should be formatted.**

Meta Prompting

Meta-prompting is a prompting strategy that emphasizes the procedural structure, reasoning pattern, or methodological framework the model should apply, rather than the specific content of an individual task.

Instead of relying on multiple task-specific examples, a meta-prompt specifies a general approach that guides the model in how to analyze, decompose, and solve a given problem.

[Zhang et al., 2024](#)

You are analyzing a short poem.

Follow this method:

1. First, explain the literal meaning of the poem in simple language.
2. Then identify the two central symbols.
3. Explain what each symbol may represent emotionally or psychologically.
4. Compare the two symbols: how are they similar, and how are they different?
5. Infer the poem's deeper message about human nature.
6. End with a short, classroom-friendly interpretation.

Poem:

“Some say the world will end in fire,
Some say in ice.
From what I’ve tasted of desire
I hold with those who favor fire.
But if it had to perish twice,
I think I know enough of hate
To say that for destruction ice
Is also great
And would suffice.”

Keep the answer clear, concise, and suitable for students new to poetry.

Meta Prompting

1. Literal Meaning:

The poem starts with the idea that some people think the world will end in fire, while others think it will end in ice. The speaker says that based on their own experiences with strong desires, they tend to agree with those who think fire is the way the world will end. However, if the world were to end a second time, the speaker knows enough about hate to realize that ice would be just as effective at destroying it.

2. Central Symbols:

The two central symbols in this poem are:

- **Fire**
- **Ice**

3. Emotional/Psychological Representation:

- **Fire:** Fire represents intense emotions, especially passion, desire, lust, and rage. It's a powerful, consuming force that can lead to rapid destruction. Think of it as burning with ambition or consumed by jealousy.
- **Ice:** Ice represents colder emotions like hatred, bitterness, apathy, and indifference. It's a slow, creeping force that can be just as destructive, but in a different way. Think of it as a deep-seated resentment or a chilling lack of empathy.

4. Comparison: Similarities and Differences:

- **Similarity:** Both fire and ice are presented as forces of destruction. They are both capable of bringing about the end of the world. They both have the potential to be all-consuming.
- **Difference:** Fire is quick, passionate, and intense. Ice is slow, calculated, and cold. Fire destroys through burning and consuming, while ice destroys through freezing and numbing. Fire is active and explosive, while ice is passive and insidious.

5. Deeper Message about Human Nature:

The poem suggests that human emotions, both the fiery passions and the icy hatreds, are powerful forces that can lead to destruction. It highlights the idea that even seemingly "cold" emotions like hate can be just as dangerous as intense emotions like desire. It implies that human nature is capable of great extremes and that both ends of the emotional spectrum can be destructive if left unchecked.

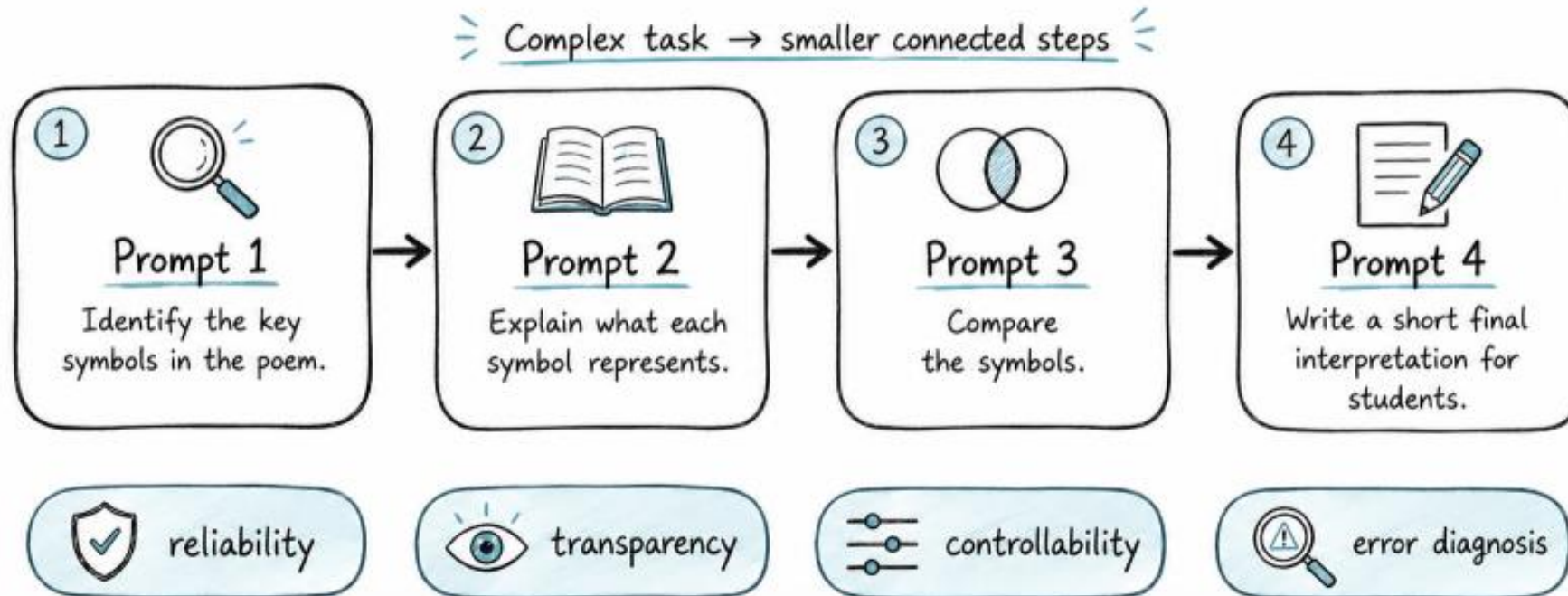
6. Classroom-Friendly Interpretation:

This poem is a warning about the power of our emotions. It tells us that both the things we desire most and the things we hate most can lead to our downfall. Whether it's a burning passion or a cold indifference, unchecked emotions can have devastating consequences. The poem reminds us that even the quietest, coldest emotions can be just as destructive as the most fiery ones.

Prompt Chaining

Prompt chaining refers to a prompting strategy in which a complex task is decomposed into a sequence of smaller, more manageable subtasks. In this approach, the output generated at one stage is used as input for a subsequent prompt, thereby creating a structured workflow of interdependent steps.

By separating the task into discrete operations, prompt chaining can enhance reliability, transparency, controllability, and error diagnosis compared with relying on a single, comprehensive prompt.



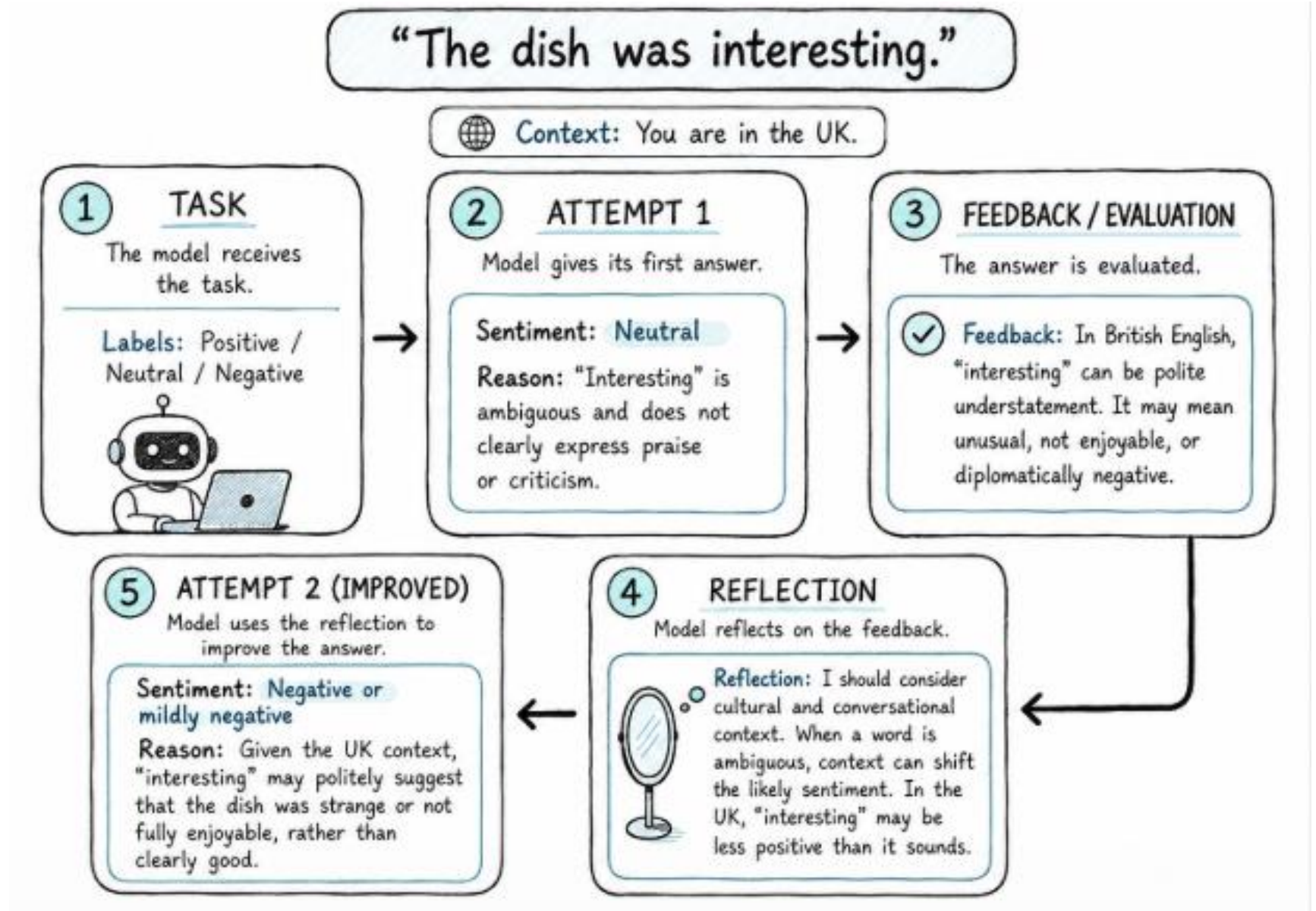
Reflexion

Reflexion is a prompting/agent strategy where an LLM not only produces an answer, but also receives feedback, reflects on what went wrong, stores that reflection, and uses it to improve the next attempt. It is especially relevant for **LLM agents**, coding tasks, multi-step reasoning, and tool-use workflows.

[Shinn et al. \(2023\)](#)

Step	What happens
1. Task	The agent receives a goal
2. Attempt	It generates an answer or action sequence
3. Evaluation	The result is checked by tests, rules, another model, or human feedback
4. Reflection	The model writes a short diagnosis of what failed and how to improve
5. Retry	The next attempt uses the reflection as memory/context

Reflexion



Agenda

01

Session goals

02

Prompt engineering

03

Exercise

04

Prompt techniques

05

Quiz

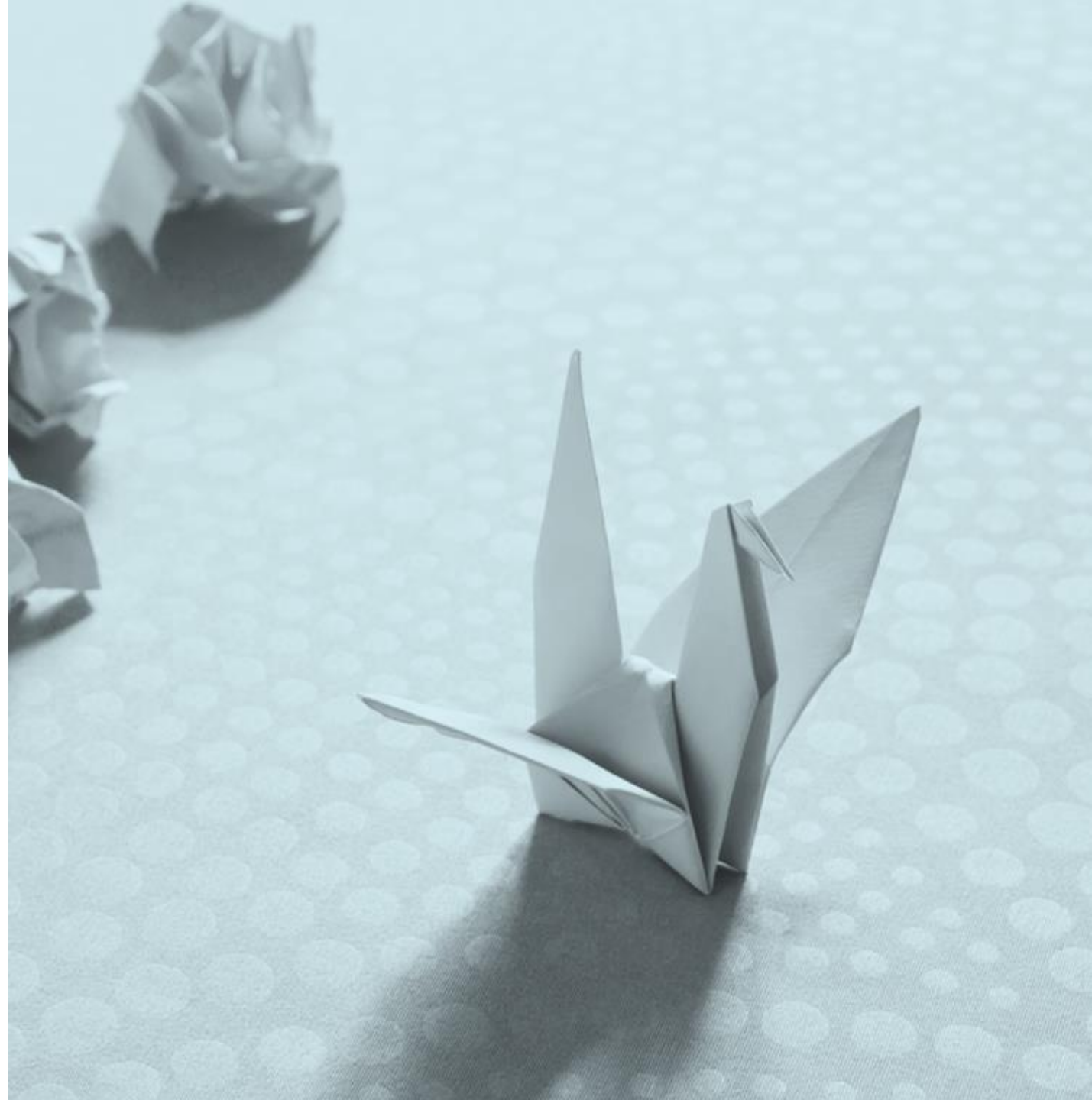
Prompt description

Your
answer

- 1 You ask an LLM: “Summarize this article in five bullet points.” You do not provide examples.
- 2 You want the model to classify customer feedback as positive, neutral, or negative. You give three example sentences with correct labels before giving the new sentence.
- 3 You ask the model to analyze a poem by first identifying symbols, then explaining their meaning, then describing tone, and finally giving a short interpretation.
- 4 You need the model to process a long report. First it should extract key facts, then group them into themes, then write an executive summary. Each step uses the previous output.
- 5 The model gives a wrong answer. You provide feedback, ask it to reflect on the mistake, and then ask it to try again with an improved answer.
- 6 You ask: “Translate this paragraph into German.” No examples or special structure are provided.
- 7 You want the model to write product descriptions in a specific company style. You provide two examples of good product descriptions, then ask it to write a new one.
- 8 You ask the model to solve a math problem by following a method: identify the known values, choose the formula, calculate step by step, and check whether the result is plausible.
- 9 You want to generate a polished email from rough notes. First the model extracts the message, then rewrites it politely, then shortens it.
- 10 The model classifies “The dish was interesting” as positive. You tell it that in British English “interesting” may be polite understatement. It reflects on this context and changes the answer to mildly negative.
- 11 You ask the model: “Explain quantum computing to a 12-year-old.” You give no examples.
- 12 You want the model to answer legal intake questions in a very specific format. You provide several sample questions and ideal answers before giving the real question.
- 13 You ask the model not only to answer, but to first determine what kind of task it is, select an appropriate strategy, apply the strategy, and then check the result.
- 14 You are building an AI workflow where one prompt extracts deadlines, another assigns responsible persons, and a final prompt creates a project plan.
- 15 You ask the model to improve its own first attempt after receiving criticism from a human evaluator.

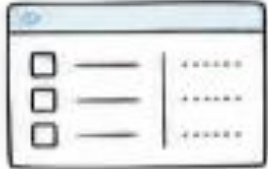
- | | |
|---|---------------------|
| A | Zero-shot prompting |
| B | Few-shot prompting |
| C | Meta-prompting |
| D | Prompt chaining |
| E | Reflexion |

Prompting quiz



From Named Techniques to Practical Prompting Habits

1 Start with a framework



Give the model a clear structure or recipe to follow.

2 Pick an expert



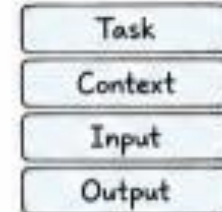
Assign a role or perspective when it helps.

3 Keep it simple (caveman prompting)



Use direct, low-noise instructions when speed matters.

4 Structure the prompt



Separate task, context, input, and output.

5 Specify the output



Define format, labels, length, and constraints.

6 Document first



Ground the answer in the source before generating.

7 Use stepwise reasoning



Break complex work into smaller steps.

- least-to-most
- chain-of-density

8 Allow uncertainty



Let the model say "I don't know" when the evidence is missing.

Framework

Instead of “Write X”, start by giving it a framework



Bad prompt: “Write an essay about automation.”

Title: [Insert Here]

Thesis: [Main Argument]

Arguments:

- [Key Point #1]
- [Key Point #2]
- [Key Point #3]

Counter-arguments:

- [Opposing View #1]
- [Opposing View #2]

Conclusion: [Wrap-up Thought]

Let AI reference or write the Framework first

Choose a well-known and effective framework for writing song lyrics, like how John Truby’s Anatomy of a Story is great for writing fiction. Describe the framework in detail in such a way that it could be used as a guide to write amazing song lyrics.

Repetition

For non-reasoning models, repeating the input prompt improves performance for popular models (Gemini, GPT, Claude, and DeepSeek) without increasing the number of generated tokens or latency.

Useful: classification, extraction

Not useful: long prompts, reasoning models

Prompt Repetition Improves Non-Reasoning LLMs

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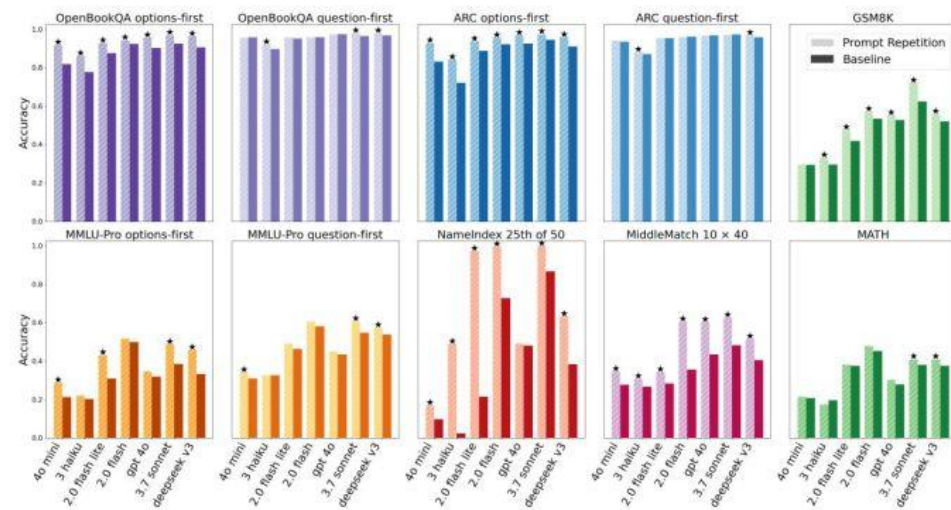
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Abstract

When not using reasoning, repeating the input prompt improves performance for popular models (Gemini, GPT, Claude, and Deepseek) without increasing the number of generated tokens or latency.

1 Prompt Repetition

LLMs are often trained as *causal* language models, i.e. past tokens cannot attend to future tokens. Therefore, the order of the tokens in a user's query can affect prediction performance. For example, a query of the form "<CONTEXT> <QUESTION>" often performs differently from a query of the form "<QUESTION> <CONTEXT>" (see *options-first* vs. *question-first* in Figure 1). We propose to *repeat the prompt*, i.e. transform the input from "<QUERY>" to "<QUERY><QUERY>". This enables each prompt token to attend to every other prompt token, addressing the above. When not using reasoning, **prompt repetition improves the performance of LLMs** (Figure 1) **without increasing the lengths of the generated outputs or latency** (Figures 2 and 3).



Terse Prompting

Terse (caveman) prompting reduces output by removing low-value language. It saves tokens when brevity matters, but it can damage clarity, nuance, and reasoning if overused.

Please read the following project update and summarize the main decisions, risks, and next steps. Please make it concise and easy to understand.

Project update:
[paste text]

Answer in dense mode.

Rules:

- No greeting.
- No filler.
- No full prose unless needed.
- Keep only decision-relevant information.
- Use bullets.
- Use this format exactly:

Decisions:

- ...

Risks:

- ...

Open questions:

- ...

Next steps: Owner → action → deadline

Project update: [paste text]

Decisions:

- Launch moved to 15 June.
- Legal review required before external release.

Risks:

- API latency still unstable.
- No confirmed owner for monitoring setup.

Open questions:

- Who approves production access?
- Is fallback process documented?

Next steps:

- Maria → confirm legal review date → Friday
- Dev team → provide latency test results → Tuesday
- Product owner → assign monitoring owner → today

Terse Prompting

LLMs often default to polite, explanatory prose: greetings, hedging, transitions, and repeated framing. Terse prompting tells the model that these tokens are unwanted, so it selects shorter, denser continuations instead.

- 1. Style steering:** The model selects shorter words, bullets, fragments, and direct labels because the prompt makes that style more likely.
- 2. Reduced filler:** Greetings, hedging, transitions, and repeated framing are suppressed. These are often useful socially, but not always useful informationally.
- 3. Clear output constraints**

Useful

Useful for summaries, extraction, quick comparisons, coding fixes, logs, structured notes, and high-volume API tasks where cost and speed matter.

Not useful

Not useful for teaching, nuanced analysis, sensitive topics, customer-facing text, or tasks where explanation, tone, caveats, and reasoning quality matter.

Structure

Separate the **instruction**, **context**, **input**, and **output format**. This helps the model parse what is task, what is data, and what is the expected answer. OpenAI, Google, and Anthropic all recommend clear structure, labels, and delimiters such as " " ", ###, or XML-style tags.

Task:

Classify the sentiment.

Labels:

Positive, Neutral, Negative

Text:

""

The dish was interesting.

""

Output:

Sentiment: <one label>

Reason: <one short sentence>

Specify Output Contract

Do not just ask for an answer. Tell the model the **format, length, style, and allowed labels**. This improves consistency, especially if the answer will be used in slides, spreadsheets, JSON, or automated workflows. OpenAI and Google both emphasize being specific about the desired output and using explicit formatting instructions.

Return exactly this format:

Sentiment: Positive | Neutral | Negative

Confidence: High | Medium | Low

Reason: one sentence only

Document First

For long documents, Anthropic recommends putting long-form data near the top, using XML tags for document structure, and placing queries at the end; they also suggest asking the model to extract relevant quotes before answering.

```
<document>
```

```
[paste long report]
```

```
</document>
```

```
<task>
```

```
First extract 3 relevant quotes from the document.
```

```
Then answer the question using only those quotes.
```

```
</task>
```

```
<question>
```

```
What are the main risks for the project?
```

```
</question>
```


Least-to-Most Prompting

Use this when a problem is too complex to solve directly. Ask the model first to break the task into simpler subproblems, then solve them in order. The original paper describes least-to-most prompting as decomposing a complex problem into simpler subproblems and solving them sequentially.

Solve this problem using least-to-most prompting.

Step 1: Break the task into simpler subquestions.

Step 2: Solve each subquestion in order.

Step 3: Use the previous answers to produce the final answer.

Problem:
[complex task]

LEAST-TO-MOST PROMPTING ENABLES COMPLEX REASONING IN LARGE LANGUAGE MODELS

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ABSTRACT

Chain-of-thought prompting has demonstrated remarkable performance on various natural language reasoning tasks. However, it tends to perform poorly on tasks which requires solving problems harder than the exemplars shown in the prompts. To overcome this challenge of easy-to-hard generalization, we propose a novel prompting strategy, *least-to-most prompting*. The key idea in this strategy is to break down a complex problem into a series of simpler subproblems and then solve them in sequence. Solving each subproblem is facilitated by the answers to previously solved subproblems. Our experimental results on tasks related to symbolic manipulation, compositional generalization, and math reasoning reveal that least-to-most prompting is capable of generalizing to more difficult problems than those seen in the prompts. A notable finding is that when the GPT-3 `code-davinci-002` model is used with least-to-most prompting, it can solve the compositional generalization benchmark SCAN in any split (including length split) with an accuracy of at least 99% using just 14 exemplars, compared to only 16% accuracy with chain-of-thought prompting. This is particularly noteworthy because neural-symbolic models in the literature that specialize in solving SCAN are trained on the entire training set containing over 15,000 examples. We have included prompts for all the tasks in the Appendix.

1 INTRODUCTION

Despite the great success of deep learning in the past decade, there still remain huge differences between human intelligence and machine learning: (1) Given a new task, humans usually can learn to accomplish it from only a few demonstration examples, while machine learning requires a large amount of labeled data for model training; (2) Humans can clearly explain the underlying rationale for their predictions or decisions, while machine learning is essentially a black box; (3) Humans can solve problems more difficult than any they have seen before, while for machine learning, examples in training and testing are typically at the same level of difficulty.

The recently proposed chain-of-thought prompting approach (Wei et al., 2022; Chowdhery et al., 2022) has taken a significant step for narrowing the gap between human intelligence and machine intelligence. It combines the idea of natural language rationales (Ling et al., 2017; Cobbe et al., 2021) with few-shot prompting (Brown et al., 2020). When further integrated with self-consistency decoding (Wang et al., 2022b) rather than using the typical greedy decoding, few-shot chain-of-thought prompting largely outperforms the state-of-the-art results in the literature on many challenging natural language processing tasks obtained from specially designed neural models trained with hundreds of times more annotated examples, while being fully interpretable.

However, chain-of-thought prompting has a key limitation—it often performs poorly on tasks that require generalization of solving problems harder than the demonstration examples, such as compositional generalization (Lake & Baroni, 2018; Keysers et al., 2020). To tackle such easy-to-hard generalization issues, we propose *least-to-most prompting*. It consists of two stages: first decomposing a complex problem into a list of easier subproblems, and then sequentially solving these subproblems, whereby solving a given subproblem is facilitated by the answers to previously solved

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Chain-of-Density

Useful when summaries are too vague. Start with a short summary, then iteratively add missing important entities without increasing length. The Chain-of-Density paper describes generating an initial sparse summary and then adding missing salient entities while keeping the length constant.

Summarize the text in 80 words.

Then revise it twice.

Each revision must:

- keep the same length
- add missing important entities
- remove vague phrases
- improve information density

Text:

[paste article]

From Sparse to Dense: GPT-4 Summarization with Chain of Density Prompting

Griffin Adams, Alex Fabbri, Faisal Ladhak, Eric Lehman, Noémie Elhadad

Abstract

Selecting the “right” amount of information to include in a summary is a difficult task. A good summary should be detailed and entity-centric without being overly dense and hard to follow. To better understand this tradeoff, we solicit increasingly dense GPT-4 summaries with what we refer to as a “Chain of Density” (CoD) prompt. Specifically, GPT-4 generates an initial entity-sparse summary before iteratively incorporating missing salient entities without increasing the length. Summaries generated by CoD are more abstractive, exhibit more fusion, and have less of a lead bias than GPT-4 summaries generated by a vanilla prompt. We conduct a human preference study on 100 CNN DailyMail articles and find that humans prefer GPT-4 summaries that are more dense than those generated by a vanilla prompt and almost as dense as human written summaries. Qualitative analysis supports the notion that there exists a tradeoff between informativeness and readability. 500 annotated CoD summaries, as well as an extra 5,000 unannotated summaries, are freely available on HuggingFace (https://huggingface.co/datasets/griffin/chain_of_density).

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Allow the Model to Say 'No'

Do not bury critical information in the middle of a huge prompt. Long-context research shows that models often perform best when relevant information is near the beginning or end, and worse when it is buried in the middle. Ask the model to explicitly state when something is 'not found'.

Important instruction:

Use only the information in the context. If the answer is not there, say "not found."

Context:

"""

[long document]

"""

Question:

Based on the context above, what is the deadline?

JSONify

JSONify prompting means expressing the task, constraints, inputs, and desired output as a structured JSON object rather than as an open-ended natural-language request. This technique is useful when the model should follow a precise specification, especially for classification, extraction, automation, API workflows, or outputs that need to be reused by machines. It helps reduce ambiguity by making the prompt components explicit: what the task is, what data should be processed, which constraints apply, and what format the answer should follow.

"Give me a deep analysis of all dentists in my area"

```
{  
  
  "task": "research",  
  
  "topic": "dentists",  
  
  "location": "DFW",  
  
  "output_format": "excel",  
  
  "tone": "professional",  
  
  "entries": 450  
}
```

Use JSONify when:

you need structure, consistency, machine-readable output, or integration into a workflow.

Avoid JSONify when:

the task is exploratory, creative, conversational, emotionally sensitive, or when a simple natural-language prompt is sufficient.